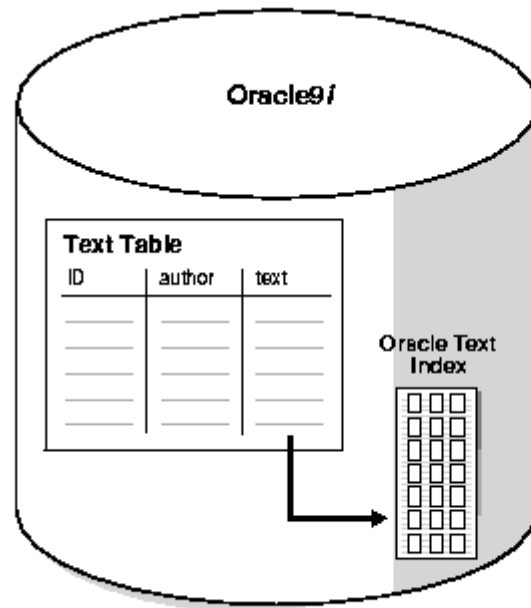


UNIT III: 3.6 – 3.7 INDEXES: Index Types, Creating an Index: Simple Unique and Composite Index, Dropping Indexes



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Objectives

After completing this lesson, you should be able to do the following:

- Create and maintain indexes.

INDEXES

An index:

- Is a schema object
- Improves the performance of some queries
- Is used by the Oracle server to speed up the retrieval of rows by using a pointer
- Can reduce disk I/O by using a rapid path access method to locate data quickly
- Is independent of the table that it indexes
- Is used and maintained automatically by the Oracle server

NOTE:

- If we need to access data quickly, we index it. If the performance of data access isn't a concern, we don't index it.

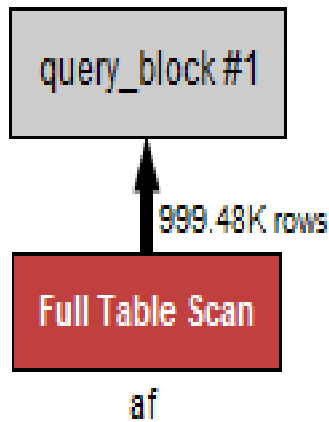
INDEXES

An index:

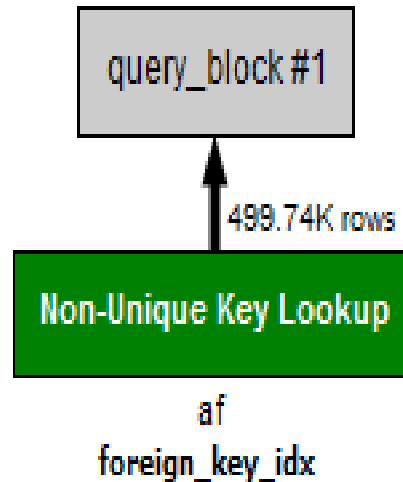
- An index in a database is very much like the index in a book.
- And, just as a book's index points to locations within the book, a database's index points to records within the storage engine.
- The database organizes the indexes such that these pointer values can be found extremely fast.
- And, once the database has located a desired set of pointer values, it can then turn around and quickly read the full table records.
- So, in the end, a database index is a means to locate and read table data as quickly as possible.

INDEXES

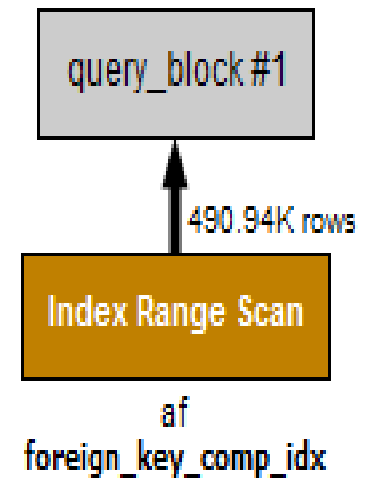
No Index



With Index



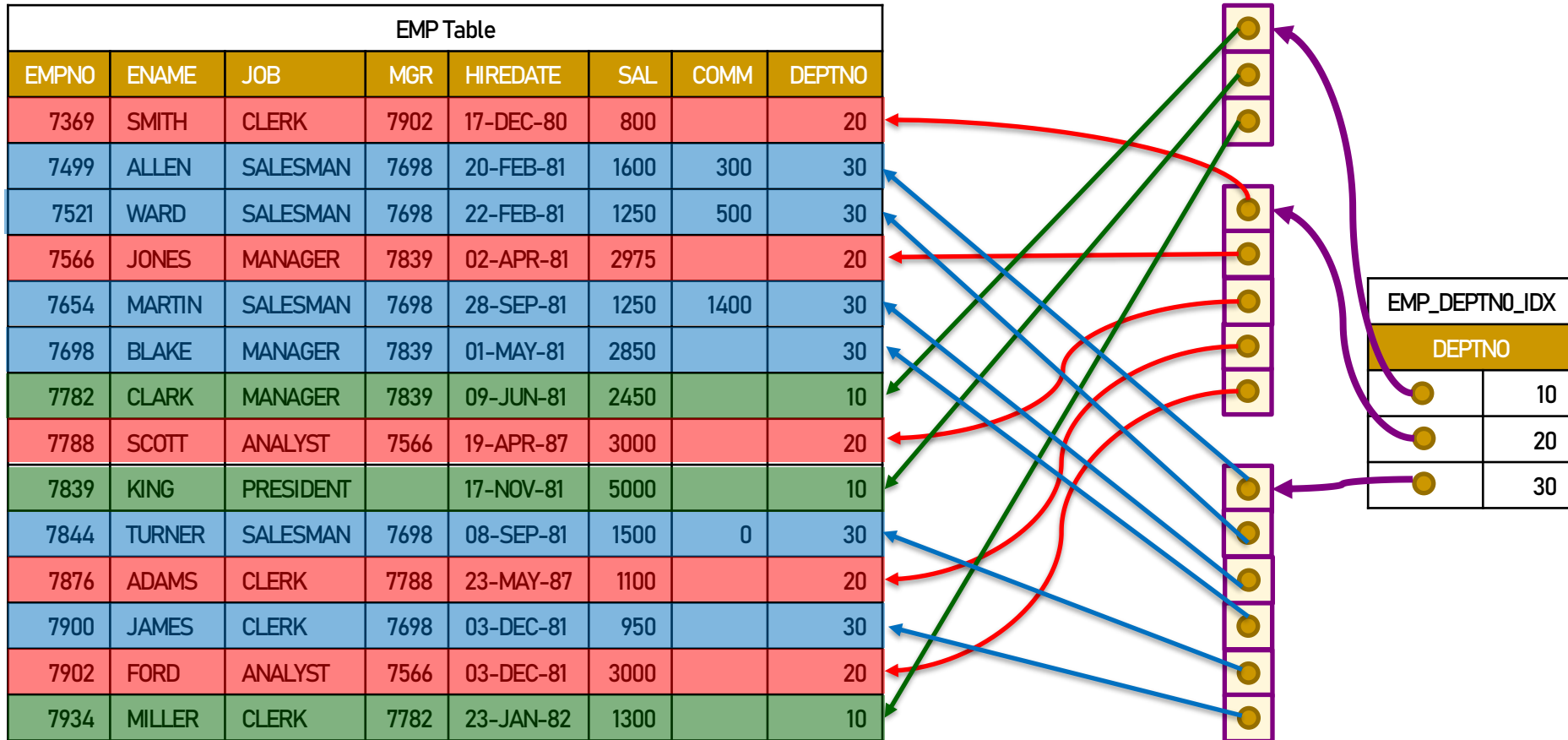
With composite Index



INDEXES – Primary

PK_EMP INDEX		EMP Table							
EMPNO		EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369		7369	SMITH	CLERK	7902	17-DEC-80	800		20
7499		7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300	30
7521		7521	WARD	SALESMAN	7698	22-FEB-81	1250	500	30
7566		7566	JONES	MANAGER	7839	02-APR-81	2975		20
7654		7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
7698		7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7782		7782	CLARK	MANAGER	7839	09-JUN-81	2450		10
7788		7788	SCOTT	ANALYST	7566	19-APR-87	3000		20
7839		7839	KING	PRESIDENT		17-NOV-81	5000		10
7844		7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30
7876		7876	ADAMS	CLERK	7788	23-MAY-87	1100		20
7900		7900	JAMES	CLERK	7698	03-DEC-81	950		30
7902		7902	FORD	ANALYST	7566	03-DEC-81	3000		20
7934		7934	MILLER	CLERK	7782	23-JAN-82	1300		10

INDEXES – Secondary



Secondary index on deptno field of EMP table.

Index record points to a bucket that contains pointers to all the actual records with that particular search-key value.

7 Secondary Indices have to be dense.

Creating INDEXES

Indexes can be created in two ways:

- **Automatically:** A unique index is created automatically when we define a **PRIMARY KEY** or **UNIQUE** constraint in a table definition.



- **Manually:** Users can create nonunique indexes on columns to speed up access to the rows.



Creating an INDEX – Simple Index

- To create an index on one column:

```
CREATE INDEX <index_name>  
ON <table_name> (column);
```

- Duplicate values are allowed.
- To improve the speed of query access to the **ENAME** column in the **EMP** table:

```
CREATE INDEX emp_ename_idx  
ON emp(ename);
```

```
Index created.
```

Creating an INDEX – Unique Index

- To create an index on one or more columns:

```
CREATE UNIQUE INDEX <index_name>  
ON <table_name> (column[, column]...);
```

- An index can be unique or non-unique. A unique index ensures that no two rows of a table have duplicate values in the indexed column (or columns).
- The following example creates a unique index on the dname column of dept table:

```
CREATE UNIQUE INDEX dept_dname_idx  
ON dept(dname);
```

```
Index created.
```

Creating an INDEX – Composite Index

- To create an index on multiple columns:

```
CREATE INDEX <index_name>  
ON <table_name> (column[, column]...);
```

- Index can be created on multiple columns in a table.
- Duplicate values are allowed.
- The following example creates a composite or concatenated index on the ENAME and JOB column of EMP table:

```
CREATE INDEX emp_job_idx  
ON emp(ename, job);
```

```
Index created.
```

Creating an INDEX – Guidelines

Create an index when:

- | | |
|-------------------------------------|--|
| ☒ | A column contains a wide range of values |
| ☒ | A column contains a large number of null values |
| ☒ | One or more columns are frequently used together in a WHERE clause or a join condition |
| ☒ | The table is large and most queries are expected to retrieve less than 2% to 4% of the rows in the table |

Do not create an index when:

- | | |
|--------------------------|---|
| ☐ | The columns are not often used as a condition in the query |
| ☐ | The table is small or most queries are expected to retrieve more than 2% to 4% of the rows in the table |
| ☐ | The table is updated frequently |
| ☐ | The indexed columns are referenced as part of an expression |

Removing an INDEX

- We can remove an index from the data dictionary by using the **DROP INDEX** command:

```
DROP INDEX <index_name>;
```

- To remove the **EMP_ENAME_IDX** index from the data dictionary :

```
DROP INDEX emp_ename_idx;
```

```
Index created.
```

- To drop an index, we must be the owner of the index or have the **DROP ANY INDEX** privilege.

A large, stylized graphic featuring the words "Question & Answer" in a bold, sans-serif font. The word "Question" is in blue, and the ampersand "&" is in yellow. The word "Answer" is also in blue. The text is centered within a large, horizontal oval shape. The oval has a thick yellow border with a blue outline. The background is white.

Question & Answer